

Apratim Shukla

Davis, California, USA

aprshukla@ucdavis.edu | +1 530 601 8382 | [GitHub](#) | [LinkedIn](#)

Education

University of California, Davis

Master of Science in Computer Science, GPA: 3.91/4

September 2022 - June 2024

California, USA

Vellore Institute of Technology

Bachelor of Technology in Electronics and Computer Engineering, CGPA: 9.28 / 10, Top 2%

July 2018 - June 2022

Chennai, India

Coursework: Algorithms, Databases, Software Engineering, Cloud and Distributed Computing, Networks

Experience

ExpoLab, University of California, Davis

Graduate Student Researcher

December 2022 - Present

Davis, USA

- Developed ResVault, a Chrome extension for Apache ResilientDB, enabling streamlined submission and retrieval of transactions, cutting response times by 10% with **Strawberry GraphQL** and **React** enhancements.
- Developed Debitable, a multi-blockchain banking app, optimizing performance across Apache ResilientDB, Ethereum, and Radix with tailored smart contracts.
- Created and optimized a ResilientDB NFT marketplace, enhancing performance by **12%** with **Flask Asyncio** and **React**, and streamlined deployments with **GitHub Actions CI/CD**, **EC2**, **Nginx**, and **Cloudflare**.

Metal Junction Services Limited

Data Analytics Intern

August 2021 - February 2022

Kolkata, India

- Optimized server logging with **Logstash**, storing data in **Elasticsearch** for improved management.
- Leveraged **Grafana** to visualize server CPU, memory, and I/O metrics, enhancing system monitoring.
- Boosted system performance by **15%** through predictive analytics in **Kibana** log visualizations.

Center for Advanced Data Science, VIT Chennai

Research Intern

June 2021 - July 2021

Chennai, India

- Configured **Docker** on **EC2** to deploy **MySQL** databases, facilitating robust data management solutions.
- Enhanced the DSS for soil chromatography analysis by integrating JSON-based interactive visualizations in the **Electron** app, upgrading analytical capabilities.
- Streamlined soil chromatography data processes with **Python** scripting, automating extraction, manipulation, and upload, achieving a **20%** efficiency increase.

Poompuhar Shipping Corporation

Full-stack Intern

February 2021 - May 2021

Chennai, India

- Built a ferry booking app with **React**, **Express**, **MongoDB**, and **GraphQL** APIs for seamless search, booking, and ticketing.
- Co-developed an admin dashboard and secured API server with **JWT** authentication, integrating **GraphQL** for mobile app support.
- Deployed the application and API on **Digital Ocean VPS**, enhanced with **Cloudflare** for DNS, SSL, and security optimizations.

Fundboon Consulting Solutions Pvt. Ltd.

Full-stack Intern

May 2020 - January 2021

Chennai, India

- Crafted an admin dashboard enhancing loan and user management with **React**, **Express**, **MongoDB**, and **GraphQL**, streamlining operations.
- Revamped user interface, introduced OTP and loan tracking, and improved response times by **50%** with **React** and **Express**.
- Implemented CI/CD via **GitHub Actions**, deploying the web app on **Heroku** for reliable and continuous delivery.

Projects

VITask | [Web Scraping](#) | [GitHub](#)

June 2019

- Stack/Tools:** Flask, Firebase, Electron, React Native, Flutter
- Founded VITask, managing a service for **400+** users and handling over **100,000** monthly requests by leveraging Python's **BeautifulSoup** and **Requests** libraries to efficiently fetch and parse data from the VIT Chennai VTOP website.

Technical Skills

Languages: Python, C, C++, JavaScript, PHP, SQL, Verilog HDL, R, Java, Rust, Go, MATLAB

Frameworks/Libraries: BeautifulSoup, Requests, Socket.io, Tensorflow, Keras, Matplotlib, Numpy, Pandas, Scikit Learn

Web Tech/Database/Tools: HTML/CSS, Bootstrap, Flask, React, jQuery, MySQL, Docker, Git, AWS, GCP, Firebase, GraphQL